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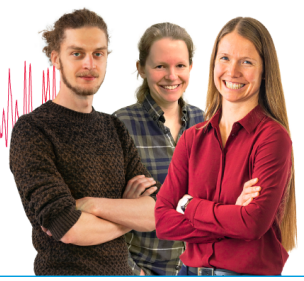
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Measurement of the vibrational state dependence of the LiH $A\ ^1\Sigma^+$ dipole moment

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Zero field molecular level crossing spectroscopy in the $J' = 2$ rotational state has been employed to measure the electric dipole moment μ_v of several vibrational levels v' of the LiH $A\ ^1\Sigma^+$ excited electronic state. The dipole moments are determined to be $\mu_1 = 1.88 \pm 0.19$ D, $\mu_2 = 1.49 \pm 0.18$ D, and $\mu_5 = 0.29 \pm 0.05$ D, where the quoted uncertainties represent 95% confidence intervals. These experimental values are found to agree rather well with vibrationally averaged dipole moments from the *ab initio* calculation by Docken and Hinze [J. Chem. Phys. 57, 4928, 4936 (1972)]. The extreme dependence of μ_v on vibrational level is striking and, on the basis of the *ab initio* results, can be ascribed to a change in the electronic structure of the LiH A state with internuclear separation.

I. INTRODUCTION

As a result of its theoretical simplicity but interesting, and sometimes unusual, properties, lithium hydride continues to be the subject of numerous experimental[1-3] and theoretical[4] studies. The present work was motivated, in part, by the MCSCF calculation of Docken and Hinze,[4(b)] in which the vibrationally averaged dipole moment μ_v of the $A\ ^1\Sigma^+$ state is found to exhibit an unusually strong dependence on vibrational level: The value of μ_v is predicted to be ~2.4 D at $v' = 0$, pass through zero near $v' = 5$, and reach ~5.7 D (with opposite sign) at $v' = 20$. This peculiar behavior, a result of the well-known avoided curve crossing in alkali halides and hydrides,[5] is due to a change in the A state charge distribution from the form Li^-H^+ at small internuclear separation R to Li^+H^- at moderately large separations and finally to neutral fragments. The avoided crossing is also manifested in the unusual shape of the A state potential curve.[1(d)] Since the calculation of Docken and Hinze[4(b)] has been found to provide values of μ_v for the ground $X\ ^1\Sigma^+$ state which agree well with experiment,[2(a),(c)] as well as a very good representation of the R dependence of the A–X electronic transition moment,[6] we have measured the dipole moment of several vibrational levels of the A state to provide a further test of this theoretical study and to learn experimentally more about the electronic structure of LiH.

II. EXPERIMENTAL

In the present experiment, zero field molecular level crossing spectroscopy[7] in an electric field[8] has been utilized to measure μ_v for the vibrational levels $v' = 1, 2$, and 5 in the $A\ ^1\Sigma^+$ state of ^{7}LiH. A supersonic molecular beam of lithium hydride was prepared[3(b)] by passing H_2 continuously through molten Li at ~1300 °K and expanding the gas mixture (total pressure ~130 Torr) through an orifice (0.03 cm diam.). After two stages of differential pumping, the beam entered a parallel-plate capacitor (plate separation 1.257 ± 0.003 cm) through a 0.24×0.95 cm slit in one plate; to prevent buildup of dielectric material, a slit was also cut in the downstream plate and allowed the beam to pass completely through. Fluorescence was excited in the LiH $(v', 0)$ progression[1] of the A–X band system in the region 382 to 362 nm with a tunable nitrogen-laser-pumped dye laser beam (pulse energy ≤ 10 μJ at the excitation zone) which was perpendicular to the molecular beam and hence electric field direction (hereafter referred to as the Z axis).

In the apparatus employed, the fluorescence detector was constrained to be perpendicular both to the exciting laser beam (defined to be the X axis) and to the Z axis. Under this circumstance, it is possible to observe only $\Delta M = \pm 2$ and not $\Delta M = \pm 1$ interferences. Accordingly, the $R1$ lines were used to excite the $J' = 2$ level, and interferences between the $M' = 0$ and $M' = \pm 2$ Zeeman levels observed. The fluorescence intensity is predicted to be of the form[7-9]

$$I/I_0 = 1 + a[1 - (1 + \tau_v^2 \omega_{02}^2)^{-1}] , \tag{1}$$

where I_0 is the fluorescence intensity at zero electric field, a is the amplitude of the level crossing signal, τ_v is the radiative lifetime of vibrational level v' of the A state, and the splitting ω_{02} is given by

$$\hbar\omega_{02} = \mu_v^2 E^2 / 21 B_v hc . \tag{2}$$

Here E is the electric field strength and B_v is the excited state rotational constant. With the laser polarized along the Y axis and both P and R decay of the excited state observed, the amplitude a is calculated to be 0.214 for X polarized detection and 0.107 with no polarizer. In some cases, the increase in fluorescence signal more than compensated for the reduction in the amplitude when the detector polarizer was removed. The only wavelength selection employed with the detector was a sharp cutoff filter to reduce the amount of scattered laser light.

III. RESULTS

Figure 1 presents our results for the $v' = 1$ and 5 vibrational levels. These data, taken with the aid of a minicomputer (New England Digital Corp.), comprise a total of several thousand laser shots per point and are normalized to constant laser energy. In addition, a comparable amount of data was taken at zero electric field to compensate for beam intensity fluctuations. The

[a)] Alfred P. Sloan Research Fellow; Camille and Henry Dreyfus Teacher–Scholar.

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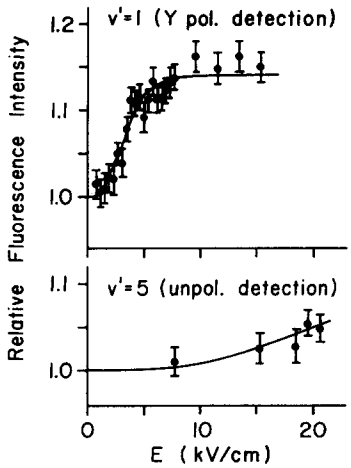


FIG. 1. Observed zero electric field level crossing signal for several LiH $A\,^1\Sigma^+$ vibrational levels. These data were taken with and without a detector polarizer for $v'=1$ and 5, respectively. Error bars represent $\pm 1\sigma$ measurement uncertainties. Solid lines represent best fits using Eqs. (1) and (2).

Stark voltages were chosen in a random manner, with zero field normalizations interspersed. The results plotted in Fig. 1 represent averages of several separate determinations at each field strength. As predicted by Eqs. (1) and (2), the signals are Lorenztians in E^2. It should be noted that the expected Stark shifts in the A–X $(v', 0)$ $R1$ line positions (~ 0.1 cm^{-1} for $E = 30$ kV/cm), which are dominated by the ground state Stark effect,[2(a),(c)] are much less than our measured laser bandwidth of 1.0 cm^{-1}, as verified in spectra vs field strength. In addition, the hyperfine splitting[2(a)] in the excited state is expected[4(b)] to be much less than the natural linewidth and hence can be ignored.

In order to determine the dipole moments, the half-width voltages $V_{1/2}$, i.e., the applied voltage at which $\tau_v \omega_{02}$ in Eq. (1) equals unity, were first determined by a nonlinear least squares fit[10] of Eq. (1) to the data. Because of the small $v' = 5$ dipole moment, it was not possible to reach the asymptotic region with the field strengths available (see Fig. 1). Accordingly, the $v' = 5$ data were fit with the amplitude a constrained to be 0.105, the predicted value with scattered laser light included in the signal.

The resulting dipole moments were then obtained from the values of $V_{1/2}$ and the plate separation (with a -4% correction[11] for the slits) with the help of experimental rotational constants[1] and radiative lifetimes.[6] For consistency, lifetimes calculated[6(d)] using experimentally derived potential energy curves and the *ab initio* A–X transition moment of Docken and Hinze[4(b)] have been used; these agree well with experimental measurements.[6(a),(b)] Table I presents the experimentally determined electric dipole moments of the ^{7}LiH $A\,^1\Sigma^+$ state. The assigned error limits include the following contributions: (1) the statistical uncertainty in the fit for $V_{1/2}$, which is the dominant factor; (2) the uncertainty in the effective spacing of the Stark plates ($\pm 1\%$); (3) the measurement accuracy of the Stark voltage ($\pm 3\%$); and (4) the uncertainty in the radiative lifetime τ_v ($\pm 5\%$). As is well known, this type of measurement does not provide the sign of the dipole moment.

IV. COMPARISON WITH THEORY

Also included in Table I are theoretically derived[4(b)] dipole moments μ_v, obtained by vibrationally averaging the dipole moment function $\mu(R)$. While μ_1 and μ_2 were not reported in Ref. 4(b), it is probably reasonable to estimate these by linear interpolation between the reported μ_0 and μ_5 values.[12] The extreme dependence of μ_v on vibrational level, in both the experimental and theoretical results, is striking. The agreement of these two sets of results with one another is very good. Indeed, for the $v' = 1$ and 2 levels it is within the experimental error.

The $v' = 5$ experimental and theoretical values differ significantly, although the absolute difference is not large. Computation of μ_5 represents a severe test for theory because the small value of μ_5 is the result of the near cancellation of two large contributions in the vibrational average of $\mu(R)$, which is varying rapidly and passing through zero near 3.5 Å. This difference could be due to inadequacies in the calculated A state potential energy curve or $\mu(R)$ function. Comparison of the former with an accurate potential energy curve[1(d)] for the $A\,^1\Sigma^+$ state shows that while vibrational energy spacings are reasonably well calculated, the $v' = 5$ turning points of the Docken and Hinze[4(b)] theoretical curve are shifted to larger R by $\delta R \sim 0.03$ Å. If the $v' = 5$ vibrational wave function were shifted by this amount, then μ_5 would change[13] by $(d\mu/dR)\delta R$ and increase in magnitude to ~ 0.2 D. Thus, the disagreement in the $v' = 5$ experimental and theoretical electric dipole moment can be attributed, at least in part, to inaccuracies in the theoretical potential energy curve. Shifting the potential energy curve would also affect the theoretical values of μ_1 and μ_2, but these would probably still lie within the experimental error limits.

Comparison of the present experimental measurement of the vibrational state dependence of the LiH A state electric dipole moment with the computation of Docken and Hinze[4(b)] suggests that their theoretical $\mu(R)$ curve is quite accurate. Hence, not only has this *ab initio* calculation predicted accurate ground X state vibrationally averaged dipole moments[2(a),(c)] and A–X electronic transition probabilities,[6] but also excited A state dipole

TABLE I. Electric dipole moments μ_v of the ^{7}LiH $A\,^1\Sigma^+$ state.

Vibrational level v'	μ_v (D) Experimental[a]	Theoretical[b]
1	1.88 ± 0.19	1.92[c]
2	1.49 ± 0.18	1.45[c]
5	0.29 ± 0.05	0.028

[a]This work. Quoted error limits represent estimated 95% confidence intervals.
[b]Reference 4(b).
[c]Linearly interpolated between the calculated μ_0 and μ_5. See Ref. 12.

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moments which agree with those measured in the present experiment.

ACKNOWLEDGMENT

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[12]Reference 4(b) reports vibrationally averaged $A\,^1\Sigma^+$ dipole moments μ_v only for every fifth vibrational level. The use of linear interpolation should be sufficiently accurate for the present comparison since the values for $v'=0$, 5, and 10 are linear with v' to within 7%. Alternatively, if a quadratic dependence of μ_v on v' is assumed, then the values of μ_1 and μ_2 are found to be 1.91 and 1.40 D, respectively, from a fit to all the μ_v reported in Ref. 4(b) ($v'=0$, 5, 10, 15, and 20). These interpolated values also lie within the experimental limits (Table I).

[13]Over the range of 3 to 4 Å, from R_e to the outer turning point, the calculated [Ref. 4(b)] A state dipole derivative $d\mu/dR$ is roughly 6 D/Å. Since the calculated $v'=5$ dipole moment is negative (Li^-H^+), shifting the vibrational wave function inward makes μ_5 more negative and hence increases its absolute magnitude.

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